

Peiru (Jenny) Xu (She/Her/Hers)

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EDUCATION

Stanford University

Stanford, CA

- Master of Science, Data Science (GPA 3.9)

Sep. 2023 – June. 2025 (Expected)

University of California, Berkeley

Berkeley, CA

- Bachelor of Arts, Applied Mathematics & Philosophy (GPA 3.9)

Aug. 2019 – May. 2023

INTERNSHIP

American Express, New York, US

Manager: Weichao Ma

Data Science Intern

June. 2024 – Aug. 2024

- Explored the best offer personalization strategy for the prospect customers using causal machine learning models
- Implemented 6 state-of-the-art uplift modeling algorithms that estimates the causal impact of different bonus offer level on the acceptance rate for card applicants, tested on multiple usecases and datasets, showing significant outperformance over baseline algorithm

Apple Inc. , Cupertino, US

Manager: Anthony Fazzi

Data Science Intern

May. 2023 – Aug. 2023

- Evaluated potential business value in current diagnostic test database, analyzed statistical correlation between customer journey and diagnostic data, and provided valuable insights to enhance data-driven decision making
- Implemented deep learning-based algorithm for image clustering, developed code base to detect newly emerging image pattern clusters based on image similarity, and achieved above 92% accuracy on test data

ABio-X Holdings, Remote/Boston, US

Manager: Prof. Mengdi Wang

Machine Learning Intern

July. 2022 – Nov. 2022

- Built and developed novel reinforcement learning algorithms to evaluate fitness of protein sequences and search for optimal new protein designs, reaching 2-fold faster convergence over baseline algorithms
- Proposed Neural Directed Evolution with TAPE embedding inputs and utilized linear bandit algorithms (LinUCB, Thompson sampling) for sequential optimization, achieving substantial improvement of performance

PROJECTS AND RESEARCH

Research Assistant, Management Science and Engineering, Stanford

Advisor: Prof. Itai Ashlagi, Prof. Irene Lo

Designing School Assignment Policy for San Francisco Unified School District

March. 2024 – June. 2024

- Modeled the student assignment system in San Francisco Unifies School District (SFUSD) to design new policies for improved diversity and less segregation
- Conducted simulations using predictive choice models and various zone-based policies with different constraints, created map visualizations of geographic and demographic data
- Created innovative new zone boundaries that prioritizes equity and diversity which largely improves socioeconomic imbalance and segregation in current SFUSD policy, expected to be implemented in SFUSD for 2026-27 school year impacting 3,000 – 5,000 students annually

Research Assistant, Department of Statistics, UC Berkeley

Advisor: Prof. Giles Hooker

Generic Model Distillation: Statistical Stability in Bayesian and Tree Model

May. 2022 – May. 2023

Zhou, Y., Xu, P., & Hooker, G. (2024). A generic approach for reproducible model distillation. *Machine Learning*, 1-44. [\[link\]](#)

- Explored the stability and interpretability of learned explanations for black-box models via generic distillation
- Developed statistical theories based on central limit theorem for determining the sample size required to guarantee stability of the distilled student model
- Conducted experiments on both simulated and real-world data to demonstrate effectiveness of the proposed generic distillation method, including tree models, falling rule list and symbolic regressions

Research Assistant, US Air Force Research Laboratory

Advisor: Dr. Alvaro Velasquez

Automaton Distillation for Non-Markovian Knowledge Transfer in Reinforcement Learning

Aug. 2021 – Dec. 2021

- Proposed automaton distillation in transfer learning to mitigate issues with traditional methods in knowledge transfer

as policy distillation

- Introduced a new and effective method of non-Markovian knowledge transfer that distills knowledge from teacher automaton to student DQN
- Experimented on a blind-craftsman environment and achieved better and more effective performance

EXPERIENCE AND SKILLS

- **Data Analysis:** Sufficient knowledge and experience in data analysis, including EDA and data visualization
Proficiency in Python, R, SQL, Spark, skillful usages of numpy, scipy, pandas, sklearn, pytorch, tensorflow
- **Machine Learning:** Coursework and projects on machine learning, including linear/logistic regression, PCA, decision trees, deep neural networks, generative models, transformers, Natural Language Processing